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20. ABSTRACT (Continue on reverse side if necessary and identify by block number)

This test operations procedure (TOP) delineates general test and specific subtest procedures required to measure the technical performance of Signal Converters. Comparison of test results with criteria and technical specifications permits evaluation of their suitability for an intended use.

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Item 20 Continued

Signal Converters are required to effect overall signaling compatibility in the many types and combinations of communication facilities, systems and associated signaling schemes comprising a military network.

This TOP is limited to tests outlined herein, which are applicable only to signal converters whose principal purpose is to alter the form of telephone or telegraph functional signals having dissimilar line or terminal signaling characteristics. Functional signals are defined as those employed in most communication circuits to operate signaling devices or switching equipment as opposed to the intelligence-bearing signals. Applicable common engineering test TOPs are listed but not included.

The following tests will measure:

a. Signaling Characteristics (Incoming Calls). The converter's ability to process a call from a 4-wire switchboard to a 2-wire subscriber and includes incoming seize and seize acknowledge, seize acknowledge delay, modularity and operator answers, subscriber release of line and outgoing release of line acknowledge.

b. Signaling Characteristics (Outgoing Calls). The converter's ability to process a call from a 2-wire subscriber to a 4-wire switchboard and includes outgoing seize signal and modularity, dial tone returned, sending DTMF digits, 4-wire subscriber-release of line and switchboard compatibility.

c. Signaling Characteristics (Release Transmission).

d. Audible Alarm. Operation of the audible alarm upon detection of a seize or release signal.

e. Tone Detectors. The operational characteristics of the tone detection system of the converter and includes seize detector sensitivity/bandwidth, release detector sensitivity/bandwidth and acknowledge detector sensitivity/bandwidth.

f. Release Detection of the converter.

g. Return Loss of the detector.

h. Transmission Loss and Frequency Response.

i. 20-Hz Rejection.

j. Longitudinal Balance.

k. Harmonic Distortion.

l. Hybrid/Transhybrid Loss.

m. Crosstalk Coupling Loss.

n. Frequency Response.

o. Noise Level of the converter.

p. System Compatibility. Converter compatibility with existing wired communication equipment.



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US ARMY TEST AND EVALUATION COMMAND
TEST OPERATIONS PROCEDURE

DRSTE -RP-702-105 10 August 1983
Test Operations Procedure 6-2-050*
AD No. A134615

# SIGNAL CONVERTERS


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1.0 SCOPE. This test operations procedure (TOP) delineates general test and specific subtest procedures for measuring and evaluating the technical performance and characteristics of Signal Converters relative to criteria specified in the Required Operational Capability (ROC), Letter Requirements (LR) and Coordinated Test Program (CTP).

The class of Signal Converters covered by this TOP are those items whose principal purpose is to alter the form of telephone or telegraph functional signals in circuits having dissimilar line or terminal signaling characteristics. Functional signals are defined as those employed in communication circuits to operate signaling devices or switching equipment as opposed to the intelligence-bearing signals (speech, data, and images).

The basic functions and characteristics of a Signal Converter are to accept information in one electrical form and relay it in a different electrical form, accurately repeat the signaling condition and sequence and provide an essentially loss-less and distortion-less intelligence-bearing signal path through the converter.

The procedures outlined in this TOP excludes the testing of items for conversion of information type signals such as multiplexers, data modems, vocoders, facsimile converters, telegraph signal converters or integral signaling components which employ out-of-band signaling.

*This TOP supersedes MTP 6-2-050, 22 September 1969.

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1.1 Common Engineering Tests. Not included in this TOP are the following Common Engineering Tests which apply to this commodity.

a. 1-2-504, Physical Characteristics

b. 1-2-609, Instructional Material Adequacy Guide and Evaluation Standard (IMAGES)

c. 1-2-610, Human Factors Engineering

d. 6-1-001, Testing Communication, Surveillance, and Avionic Electronic Equipment

e. 6-2-503, Reliability

f. 6-2-504, Maintenance/Maintainability

g. 6-2-507, Safety and Health Evaluation-Communication/Electronics

h. 6-2-514, Electrical Power Requirements

i. 6-2-521, Engineering Intelligibility Testing of Voice Communication Equipment

j. 6-2-542, Electromagnetic Interference Tests for Electronic Equipment

k. 6-4-001, Desert (Field) Environmental Testing of Communication, Surveillance, and Avionic Electronic Equipment

l. 6-4-003, Communication, Surveillance and Avionic Electronic Equipment

m. 6-4-004, Arctic Environmental Test of Tactical Radio Communication Equipment

n. 6-4-006, Arctic Environmental Test of Tactical Wire Communication Equipment

o. MIL-STD-810D, Environmental Test Methods

## 2.0 FACILITIES AND INSTRUMENTATION

2.1 Facilities. The Communication Test Facility (CTF) - The CTF is a grounded bench test facility with adjacent open areas for sites suitable for cable installation. The capability exists for the establishment of radio links as required. The CTF has immediate access to a shielded room and parts and accessories such as cabling, connectors, power supplies, tools and electronic measurement instrumentation.

2.2 Instrumentation and Equipment. Variable power supply(ies); signal generators; electronic counter (EC); wave analyzer (WA); noise measuring set (NMS); voltmeter (VM), AC; oscilloscope w/camera; audio amplifier; wavemeter; and telephone.



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2.3 Characteristics/Requirements. The characteristics and accuracies of the above test instrumentation are determined by the performance specifications of the individual signal converter to be tested and the circuits in which it is to be used. Select test equipment having an accuracy at least 10 times greater than that of the function to be measured. The above listed major facilities, instrumentation and equipment will provide the necessary characteristics and setups required to perform the following subtest procedures which will measure:

a. Signaling Characteristics (Incoming Calls). The converters ability to process a call from a 4-wire switchboard to a 2-wire subscriber and includes incoming seize and seize acknowledge, seize acknowledge delay, modularity and operator answers, subscriber release of line and outgoing release of line acknowledge.

b. Signaling Characteristics (Outgoing Calls). The converters ability to process a call from 2-wire subscriber to a 4-wire switchboard and includes outgoing seize signal and modularity, dial tone returned, sending DTMF digits, 4-wire subscriber-release of line and switchboard compatibility.

c. Signaling Characteristics (Release Transmission).

d. Audible Alarm. Operation of the audible alarm upon detection of a seize or release signal.

e. Tone Detectors. The operational characteristics of the tone detection system of the converter and includes seize detector sensitivity/bandwidth, release detector sensitivity/bandwidth and acknowledge detector sensitivity/bandwidth.

f. Release Detection of the converter.

g. Return Loss of the detector.

h. Transmission Loss and Frequency Response.

i. 20-Hz Rejection.

j. Longitudinal Balance.

k. Harmonic Distortion.

l. Hybrid/Transhybrid Loss.

m. Crosstalk Coupling Loss.

n. Frequency Response.

o. Noise Level of the converter.

p. System Compatibility. Converter compatibility with existing wired communication equipment.



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## 3.0 PREPARATION FOR TEST

a. Review TECOM Pamphlet 70-3, Research, Development, and Acquisition and Project Engineer's Handbook for guidance on test planning, execution, reporting and post-test activities.

b. Establish and/or continually maintain a readily accessible project log and project file.

c. Review local installation project office handbook, standing operating procedures (SOP) and implementing directives which govern the administrative processes of preparing test plans, conducting tests, preparing reports, reporting to the Test Resources Management System (TRMS) and budgeting.

d. Acquire and review all descriptive, instructional, and specification material on the test items issued by Government and contractor(s) for checking the test plans subtest objectives, criteria, facility(ies) and instrumentation requirements.

e. Determine the scheduled availability of the test item.

f. Ensure availability of appropriate facilities, and coordinate the test support requirements including personnel, equipment, maintenance, spare parts and instrumentation.

g. Review the detailed test plan.

h. Record as a minimum the following data:

(1) Nomenclature, serial number(s), manufacturer's name, and function of the item(s) under test.

(2) Nomenclature, serial number, accuracy tolerance, calibration requirements, and last calibration data of test equipment selected for the tests.

(3) Damages to the test item(s) incurred during transit and/or manufacturing defects.

(4) Test item photographs.

i. Establish instrumentation or measurement system mean error and standard deviation of error.

j. Determine test item sample size.



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## 4.0 TEST CONTROLS

a. Organize test team and establish responsibilities for test conduct, reporting, and data control.

b. Select test equipment having an accuracy at least 10 times greater than that of the function to be measured.

c. Familiarize all test personnel with the test item instructional material.

d. Prepare adequate safety precautions to provide safety for personnel and equipment and ensure all safety SOPs are observed throughout the test.

e. Inspect the test item thoroughly for obvious physical and electrical defects. Note and correct all defects before proceeding with testing.

f. Prepare, disseminate and continually review record forms for systematic entry of data, chronology of tests, and analysis in final evaluation of the test item.

g. Prepare and monitor a test item sample plan sufficient to ensure that enough samples of all measurements are taken to provide statistical confidence of final data in accordance with TOP 3-1-002.

h. Perform such preliminary tests as necessary to assure the test item(s) is in satisfactory condition.

## 5.0 PERFORMANCE TESTS

NOTE: Modification of these procedures shall be made as required by technical design of the test item and availability of test equipment, but only to the extent that such modified procedures will not affect the vailidity of test results.

### 5.1 Test Methods.

#### 5.1.1 Signaling Characteristics (Incoming Calls)

NOTE: This subtest is to determine if the signal converter will process a call from a 4-wire switchboard (SB-22) to a 2-wire subscriber.

##### 5.1.1.1 Incoming Seize and Seize Acknowledge

a. Select a signal converter channel for test and set up the equipment as shown in figure 1, page 6, with power supply adjusted for proper input voltage.

b. Set converter power switch to OFF, switch SW-1 to position A and the audio signal generator (OSC) to 2250 Hz at 0.31 voltage root mean square (Vrms) as read on Voltmeter VM-1.



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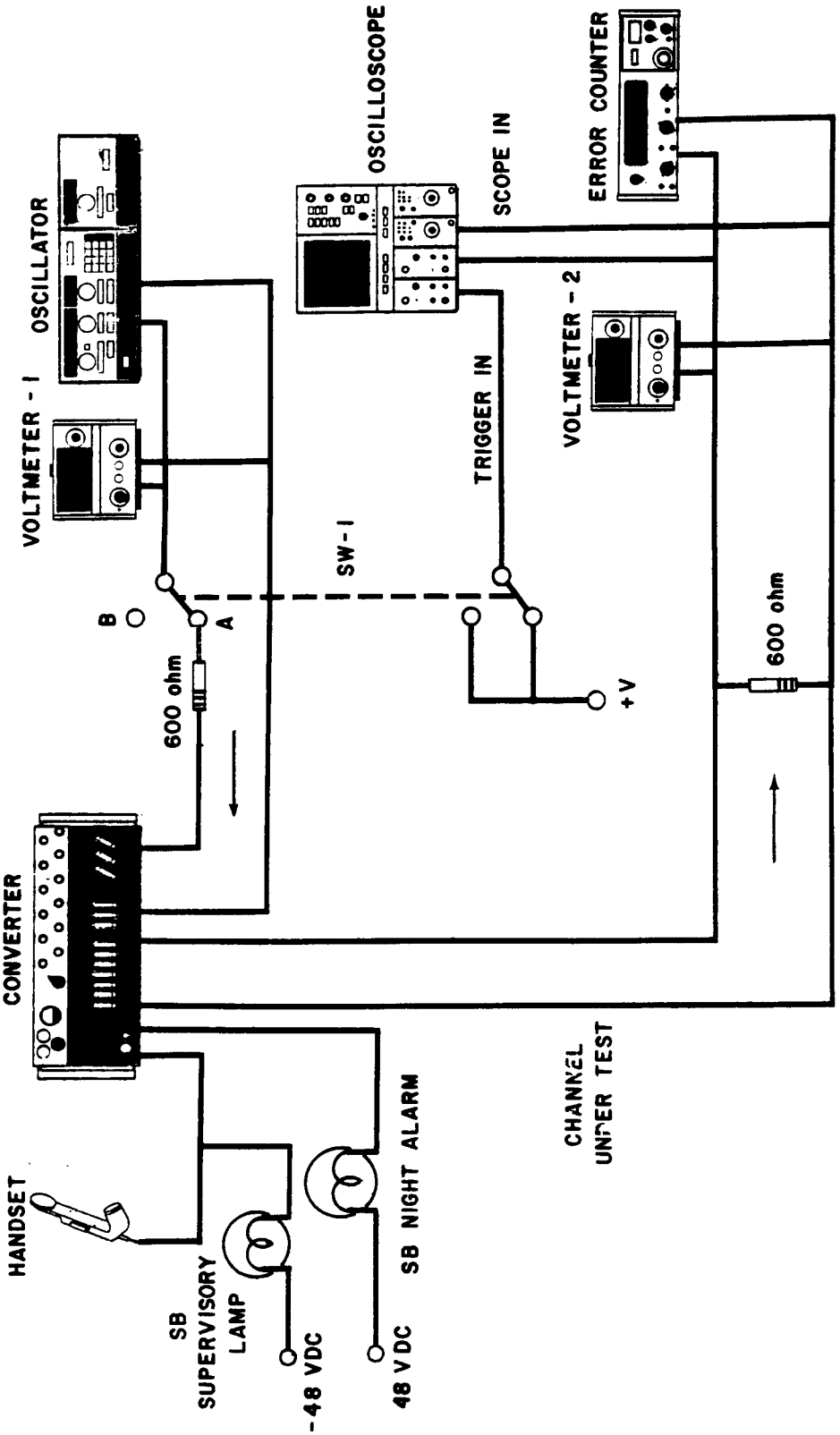



Figure 1. Incoming call test setup.



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c. Turn on the converter power switch and record the seize acknowledge tone frequency on the electronic counter and the tone level on voltmeter VM-2. Note also the condition of the night alarm and the channel under test (CUT) service request indicator (SRI). The SRl is shown in Fig. 1 as the SB Supervisory Lamp.

5.1.1.2 Seize Acknowledge Delay

a. Set up the equipment as outlined in paragraph 5.1.1.1 a above.

b. Inhibit the 425 Hz interrupter with the 425 Hz tone on continuously.

c. Place switch SW-1 in position B which will trigger the oscilloscope and stop the seize signal. Measure the delay time before the 425 Hz (ringback) signal appears on the oscilloscope. Record the ringback signal frequency from the EC and the tone level from voltmeter VM-2.

5.1.1.3 Ringback Timing

a. Refer to the set up and procedure of paragraph 5.1.1.1 above.

b. Remove the 425 Hz interrupter inhibit signal by switching SW-1 from A to B and record the ON and OFF cycle time of the ringback signal.

5.1.1.4 Modularity and Operator Answers

a. Refer to the test set up of paragraph 5.1.1.1.

b. Remove one of the converter channels not being tested from the set up and connect the switchboard operator's cord to the converter CUT.

c. Momentarily activate the converter CUT access button and record the condition of the SRI and night alarm and the frequency reading on the EC.

5.1.1.5 Call Extended to 2-wire Subscriber

a. Refer to the test set up of paragraph 5.1.1.1.

b. Extend the call at the switchboard (SB) by plugging the cord of the 2-wire party into the calling jack of the converter.

c. Set the audio signal generator (OSC) to 1000 Hz and switch SW-1 to position A.

d. Remove the handset from its cradle and note the presence or absence of the 1000 Hz tone.



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5.1.1.6 Subscriber Release of Line

a. Refer to the test set up of paragraph 5.1.1.1.

b. Place switch SW-1 in position B and activate the converter CUT access switch. Observe the oscilloscope while momentarily activating the release switch with the access switch released. Record the duration time of the 2600 Hz release signal and its frequency and level on the EC and voltmeter VM-2.

5.1.1.7 Outgoing Release of Line Acknowledge

a. Refer to the test set up of paragraph 5.1.1.1.

b. Set the audio signal generator (OSC) to 570 Hz and switch SW-1 in position A.

c. Adjust the OSC output level to 0.31 Vrms as read on voltmeter VM-1.

d. Place switch SW-1 in position B, activate and hold the CUT access switch. Observe oscilloscope for the 2600 Hz release signal while the release switch is momentarily activated and switch SW-1 immediately moved to position A. Record the delay time before the 2600 Hz signal is cut off.

5.1.2 Signaling Characteristics (Outgoing Calls)

NOTE: This subtest is to determine if the signal converter will process a call from a 2-wire subscriber to a 4-wire automatic switchboard (SB-22).

5.1.2.1 Outgoing Seize Signal and Modularity

a. Select a signal converter channel for test, remove one of the channels not under test and set up the equipment as shown in figure 2, page 9, with the power supply set to the specified voltage.

b. Set switch SW-1 to position B.

c. Insert switchboard operator's cord and calling party's cord into the CUT.

d. Activate and hold the CUT access and seize button.

e. Adjust the WA frequency control to approximately 2250 Hz and peak deflection.

f. Record the frequency and level of the seize tone on the EC and WA.

5.1.2.2 Dial Tone Returned

a. Refer to figure 2, page 9.



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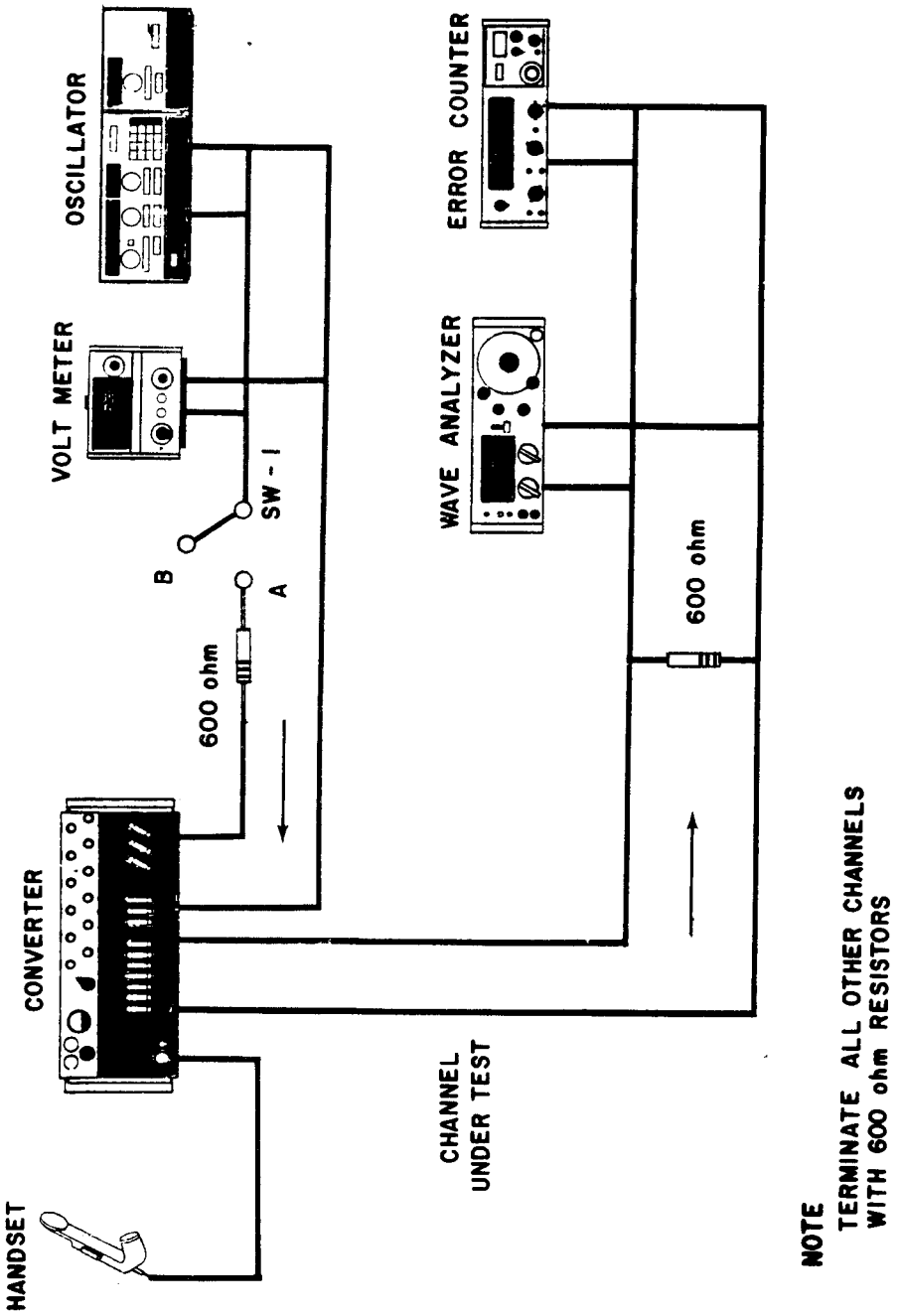



Figure 2. Outgoing call test setup.



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b. Set the audio signal generator (OSC) to 425 Hz, with its output level to 0.31 Vrms and switch SW-1 to position A.

c. Record the presence/absence of the 425 Hz (frequency and amplitude level) at the switchboard operator's position and at the handset.

5.1.2.3 Sending Dual Tone Multi-Frequency (DTMF) Digits

a. Refer to figure 2, page 9.

b. Place switch SW-1 in position B.

c. Activate and hold the CUT access button and then pushbutton 1 key call.

d. Adjust WA to approximately 697 Hz and peak deflection.

e. Record the frequency and level of the DTMF signal on the EC and WA.

f. Adjust the WA to approximately 1209 Hz and peak deflection.

g. Record the frequency and level.

h. Repeat steps a thru g for the remaining 11 key call digits. Observe the following list of frequencies:

| Pushbutton | Frequency | Pushbutton | Frequency |
|---|---|---|---|
| 2 | 697<br>1336 | 7 | 852<br>1209 |
| 3 | 697<br>1477 | 8 | 852<br>1336 |
| 4 | 770<br>1290 | 9 | 852<br>1477 |
| 5 | 770<br>1336 | 0 | 941<br>1336 |
| 6 | 770<br>1477 | C | 941<br>1477 |
| | | R | 941<br>1209 |

5.1.2.4 4-Wire Subscriber-Release of Line

a. Refer to figure 2, page 9.

b. Set the audio signal generator (OSC) to 2600 Hz at 0.31 Vrms. Switch SW-1 to position A and adjust the WA to approximately 570 Hz and peak deflection.

c. Record the frequency and level of the signal on EC and WA and the condition of the SRI.



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5.1.2.5 Switchboard-Converter Compatibility

a. Replace switchboard used in subtests of paragraph 5.1.2.1 thru 5.1.2.4 above with the other switchboard(s) (SB-86) specified for use with the signal converter under test.

b. Set the power supply (figure 2) to voltage specified.

c. Repeat steps of paragraphs 5.1.2.1 thru 5.1.2.4 above for a randomly selected signal converter channel.

5.1.3 Signaling Characteristics (Release Transmission)

a. Set up the test item and test equipment as shown in figure 3, page 12, and randomly select a channel for test.

b. Apply primary power to the converter and place switch SW-1 to position B.

c. Press and hold the channel's access switch and while observing the oscilloscope, press the release pushbutton momentarily and release the access switch.

d. Record the frequency measured on the EC and the output on voltmeter (VM-2).

e. Repeat steps b through d for the remaining converter channels.

5.1.4 Audible Alarm

a. Set up this test as shown in figure 3, select a channel for test and apply primary power to the test item.

b. Lift the hand set from its cradle and note the sounding of the night alarm.

c. Vary the volume control from maximum to OFF and place in the maximum position.

d. Reset the alarm by inserting a plug into the jack of the CUT.

e. Return the handset to ON hook and note the alarm sound. Vary the volume control from maximum to OFF and leave in the maximum position.

f. Reset the alarm by repeating step d.

g. Repeat steps d through f for the remaining converter channels.

5.1.5 Tone Detectors

a. Select a channel for test with set up of figure 3 and apply power.



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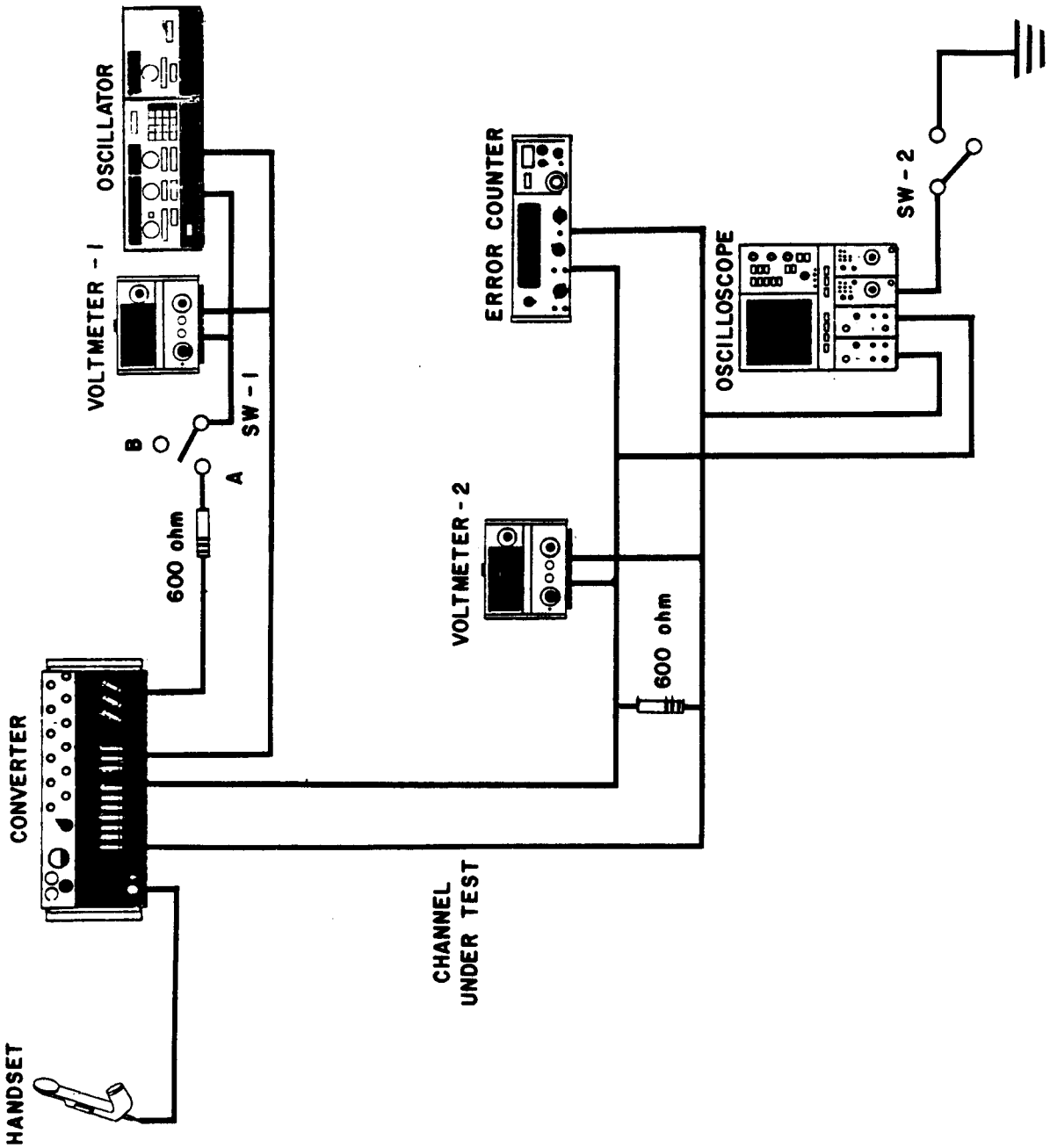



Figure 3. Seize, release, and acknowledge detectors test setup.



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b. Seize Detector Sensitivity/Bandwidth.

(1) Set the attenuator to 10 dB and the audio signal generator (OSC) to 2250 Hz at a -4 dBm level as read on the EC and voltmeter VM-2. With the attenuator set at maximum, momentarily depress the access switch to reset the SRI.

(2) Slowly decrease the attenuator until the SRI comes on and record the input level as read on voltmeter VM-1.

(3) Repeat steps (1) and (2) for audio signal generator (OSC) settings of 2217 and 2283 Hz.

(4) Repeat steps (1) and (2) for OSC settings of 2160 and 2340 where the SRI should not come on.

c. Release Detector Sensitivity/Bandwidth

(1) Insert a plug with 600 ohms into the CUT to establish a busy condition.

(2) Set the attenuator at 10 dB, the audio signal generator (OSC) to 2600 Hz with the power level adjusted to -5 dBm. Note that the night alarm and the SRI are activated.

(3) Record the voltmeter VM-1 reading in dBm.

(4) Repeat steps (2) and (3) for audio signal generator (OSC) settings of 2561 and 2639 Hz.

(5) Repeat steps (2) and (3) for OSC settings of 2496 and 2704 Hz where the SRI should not come on.

d. Acknowledge Detector Sensitivity/Bandwidth

(1) Insert a plug with 600 ohms into the CUT to establish a busy condition.

(2) Set the attenuator at 10 dB, the audio signal generator (OSC) to 570 Hz with the power level adjusted to -5 dBm. With the attenuator adjusted to maximum and the access pushbutton pressed and held, momentarily depress the release pushbutton and note the oscilloscope for presence of 2600 Hz signal.

(3) Record the length of time (3 to 10 seconds) that the 2600 Hz signal is present.

(4) Repeatedly send the release signal while slowly increasing the attenuator until the 2600 Hz signal no longer appears and record the input level of voltmeter VM-1.

(5) Repeat steps (2) through (4) for OSC settings of 547, 561 and 593 Hz.



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5.1.6 Release Detection

a. Select a converter channel for test with the test setup of figure 3, page 12, and apply primary power.

b. Adjust the audio signal generator (OSC) to 570 Hz.

c. Set switch SW-1 to position A and adjust OSC output level to 0.31 Vrms as read on Voltmeter VM-1. Return switch SW-1 to position B.

d. Press and hold the channel access switch.

e. Momentarily push the release switch and immediately close switch SW-1 to position A while observing the oscilloscope where the 2600 Hz release signal should cut off in 3 to 10 seconds.

f. Repeat steps b through e for the remaining test item channels.

5.1.7 Return Loss

a. Select a converter channel for test and set up the equipment as shown in figure 4, page 15, with power supply output as specified.

b. Set the attenuator (ATT) to 20 dB, the audio signal generator (OSC) to 300 Hz, SW-1 to the SET LEVEL position and the WA mode selector switch to the NORMAL position. Adjust the WA frequency control for a peak meter reading around 300 Hz and the OSC output at a -4 dBm on the WA meter.

c. Move the WA mode selector switch on the AFC position and switch SW-1 to the MEASURE position. Record the return loss in dB.

d. Repeat steps b and c for frequencies of 475, 550, 750, 1000, 2600, 3000 and 3500 Hz.

e. Repeat steps b through d with points A and B connected to the CUT 4-wire receive terminals and the 2-wire send terminals terminated in 600 ohms.

f. Repeat steps b through d with points A and B connected to the CUT 4-wire receive terminals and the 4-wire send terminals terminated in 600 ohms.

g. Repeat steps b through f on each of the remaining test converter channels.

5.1.8 Transmission Loss

a. 2-wire to 4-wire

(1) Set up the test as depicted in figure 5, page 17, and randomly select a converter channel for test and set the power supply level at the specified voltage.



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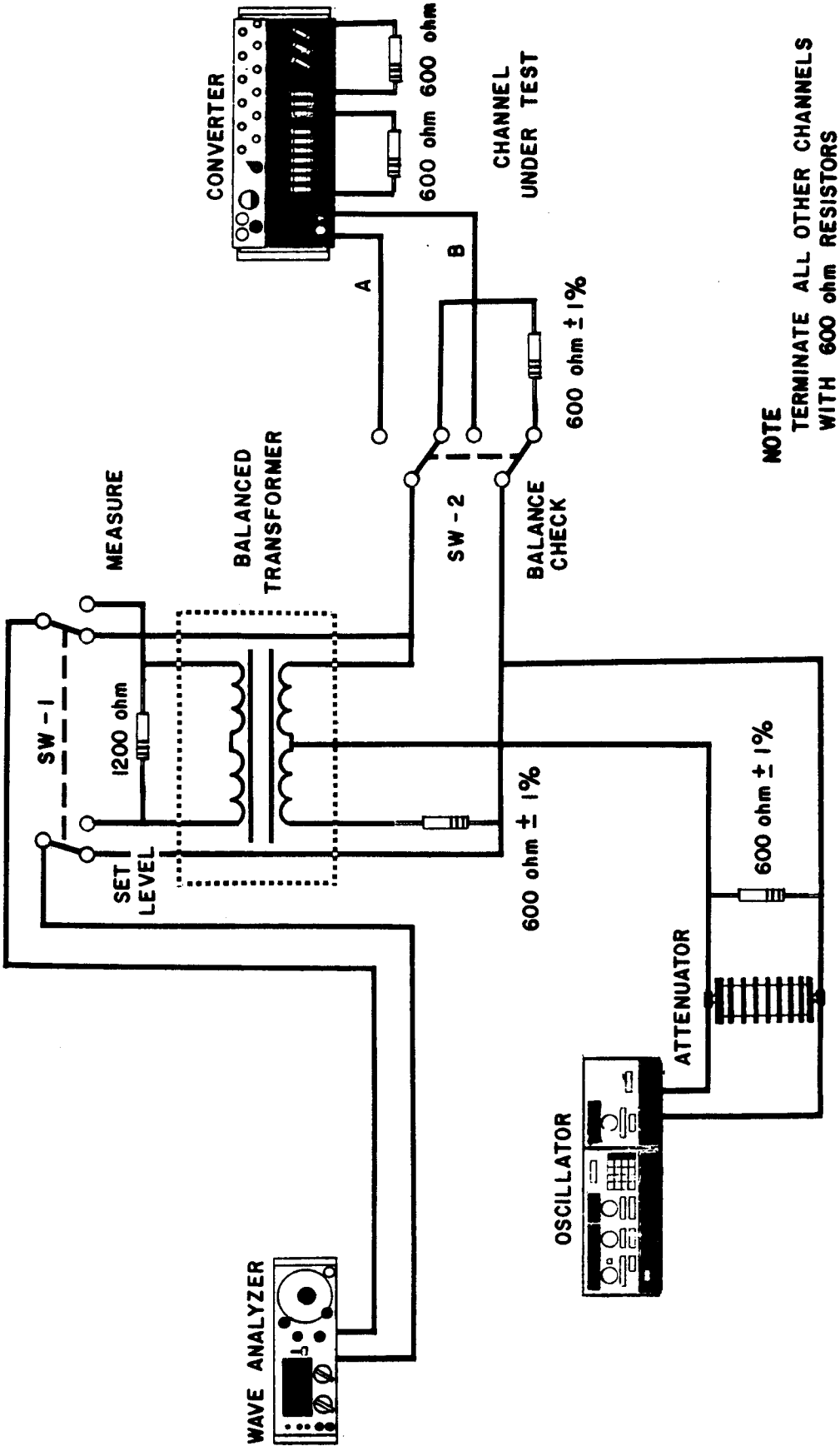



Figure 4. Return loss test setup.



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(2) Set the audio signal generator (OSC) to 300 Hz at 0.98 Vrms as read on voltmeter VM-1. Record the output in dBm as indicated on voltmeter VM-2.

(3) Repeat step (2) for test frequencies of 500, 1000, 1500, 2000, 2500, 3000 and 3500 Hz and 5, 10, 20, 30, 40, 50, 75, and 100 kHz.

b. 4-wire to 2-wire

(1) Set up the test as shown in figure 6, page 18.

(2) Repeat steps a (2) and a (3) above.

c. Repeat step a and b above for the remaining converter channels.

5.1.9 20-Hz Rejection

a. Set up the test as shown in figure 6, page 18, and randomly select a converter channel for test.

b. Set the audio signal generator (OSC) to 20 Hz and the OSC Amplifier (AMP) to 90 Vrms as read on voltmeter VM-1. Record the output level in dBm on voltmeter VM-2.

c. Repeat step b for the remaining channels.

5.1.10 Longitudinal Balance

a. Set up the test as shown in figure 7, page 19, with the power supply set at the specified voltage and a randomly selected converter channel for test.

b. Set switch SW-1 in position A and the WA (OSC) to 300 Hz at 0.98 Vrms as read on the voltmeter. Record as dB 1 the output on the WA meter.

c. Set switch SW-1 to position B and the WA OSC as in b above. Record the output on the WA meter as dB 2.

d. Calculate longitudinal balance (LB) as

LB = dB 2-dB 1

e. Repeat steps b through d for test frequencies 300, 500, 1000, 1500, 2000, 2500, and 3500 Hz.

f. Repeat steps b through e for the remaining converter channels.

5.1.11 Harmonic Distortion

a. Set up the test as shown in figure 8, page 21 with a randomly selected converter channel.



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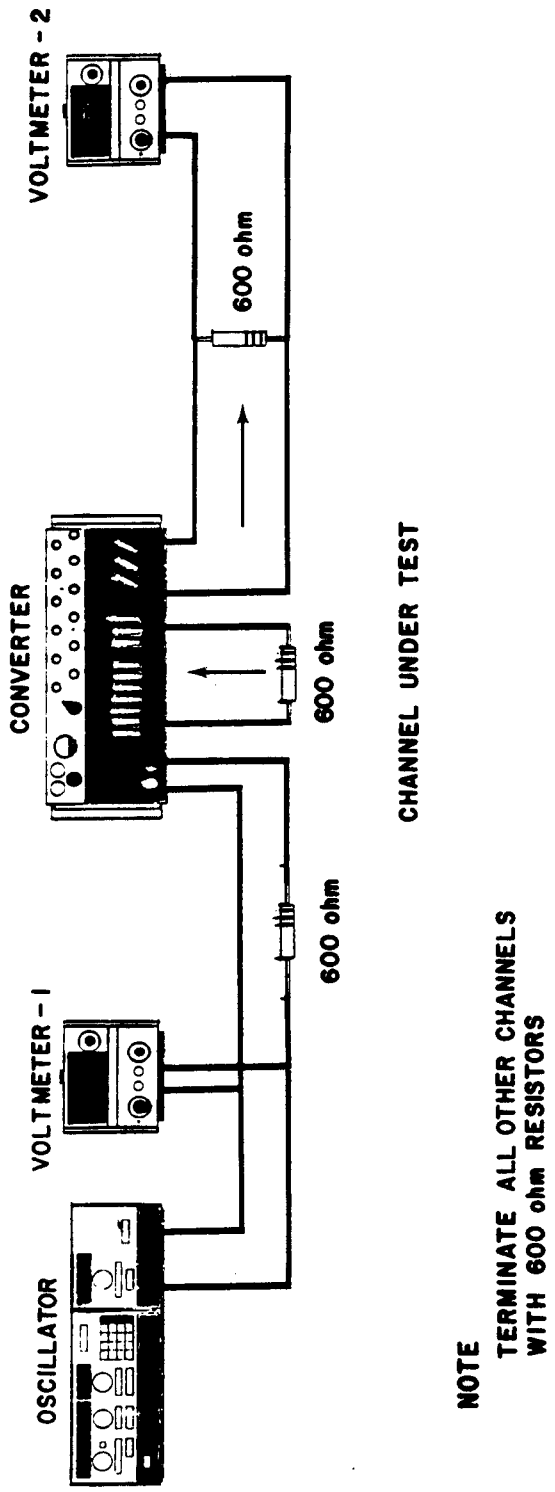



Figure 5. Transmission loss test setup.



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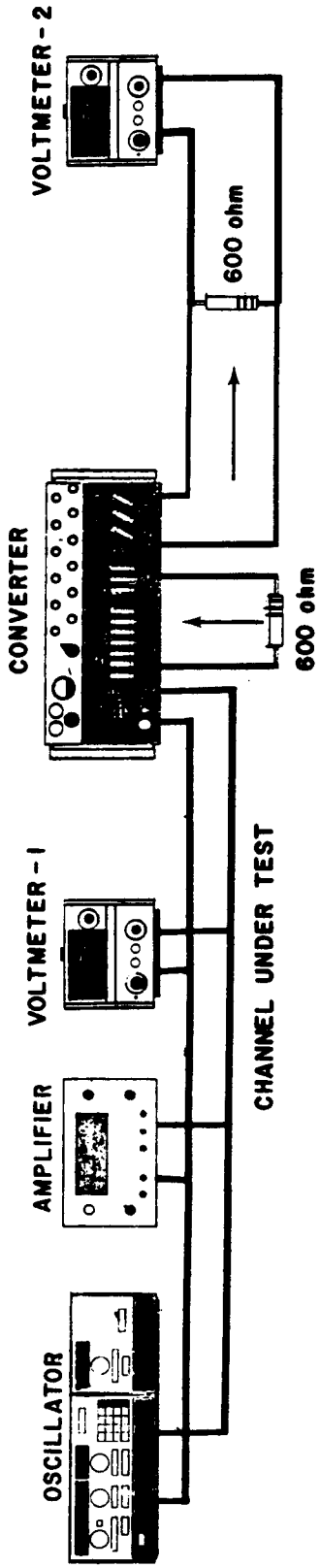



Figure 6. Rejection loss test setup.



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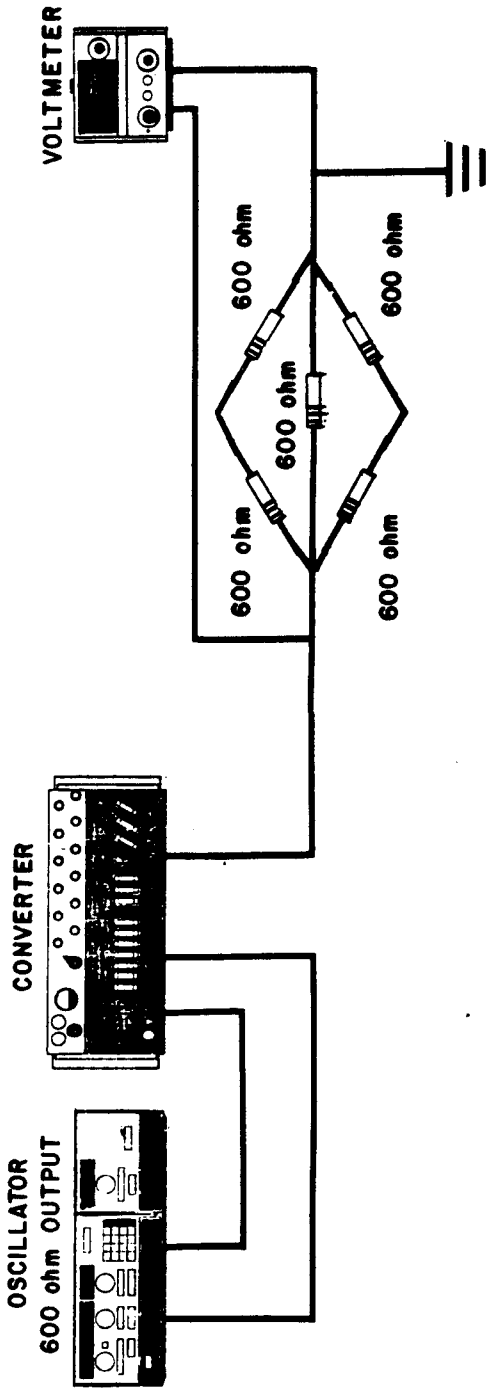



Figure 7. Longitudinal balance test setup.



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b. Set up the audio signal generator (OSC) to 300 Hz at 0.98 Vrms (fundamental). Record the levels of the first through tenth harmonics as read on the WA meter.

c. Repeat step b for fundamental frequencies of 500, 1000, 1500, 2000, 2500, 3000, and 3500 Hz.

5.1.12 Hybrid/Transhybrid Loss

a. Set up the test as shown in figure 9, page 22, with a randomly selected converter channel and the power supply set at the specified voltage.

b. Disable the converter CUT limiter [by placing an appropriate value (approx, 47-mfd) capacitor from proper test point on CUT to test point on the common module].

c. Place both input and output attenuator ATT switches to the ATT position and set the ATT to 4 dB.

d. Set the WA input voltage to 1 volt, the scale value to absolute and the range switch to -10 dBm. Set the limits of the sweep drive to sweep between 200 and 4000 Hz.

e. Adjust the WA output until +2 dB is read on the meter with the frequency set at 1 kHz.

f. Plot the curve of -30, -35, and -40 dB using the ATT for reference levels.

g. Plot the receive to send response of the CUT.

h. Verify the hybrid loss of the CUT to assure it is within specified limits for the frequencies of 500, 1000, 1500, 2000, 2500, 3000 and 3500 Hz.

i. Repeat steps b through h for the remaining channels.

5.1.13 Crosstalk

a. Set up the test as shown in figure 10, page 23 with a randomly selected converter channel for test and the power supply set to the specified voltage.

b. Set the audio signal generator (OSC) to 300 Hz and 0.98 Vrms.

c. Set the WA input voltage to 1 volt, scale value to absolute and the range value to 0 dBm and record WA output and voltmeter WM-2 reading.

d. Repeat steps b through d for the remaining channels.

5.1.14 Frequency Response

a. Set up the test as shown in figure 11, page 25 with a randomly selected converter channel for test.



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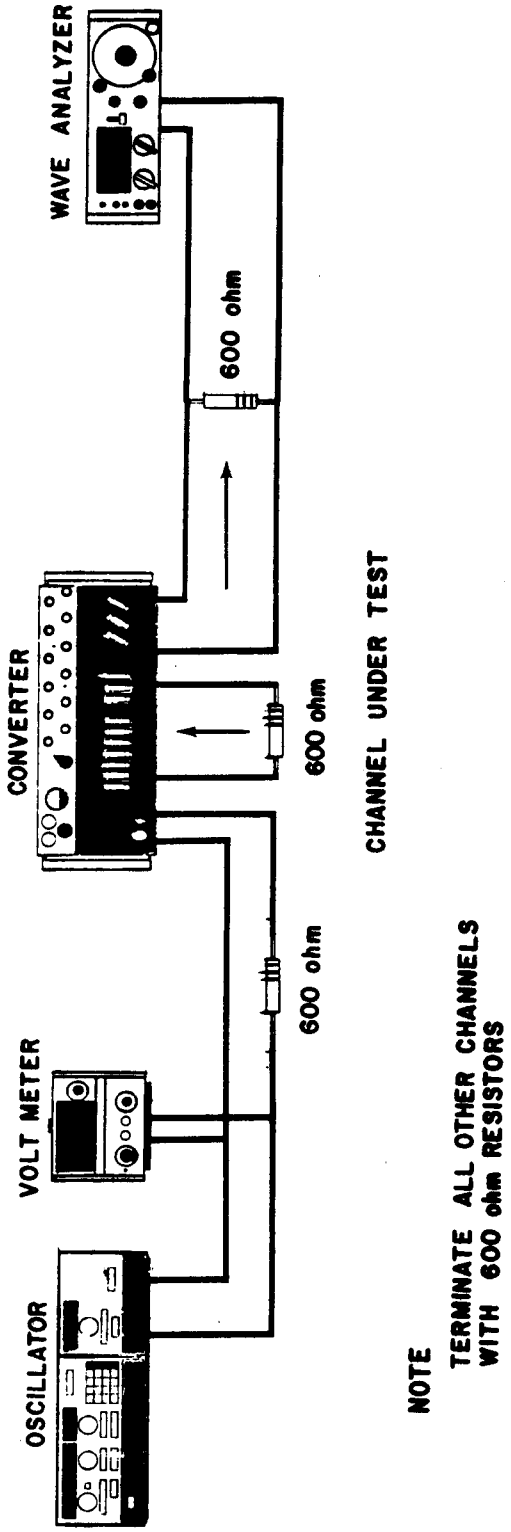



Figure 8. Harmonic distortion test setup.



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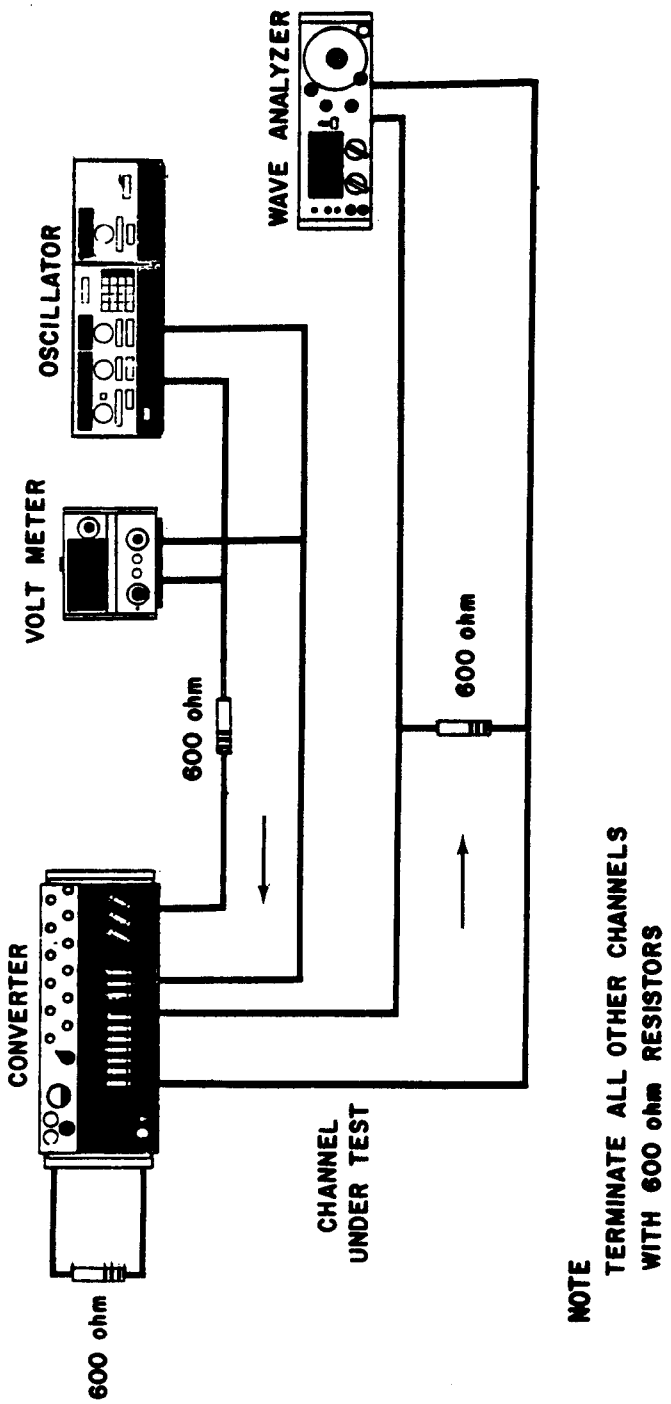



Figure 9. Hybrid/transhybrid loss test setup.



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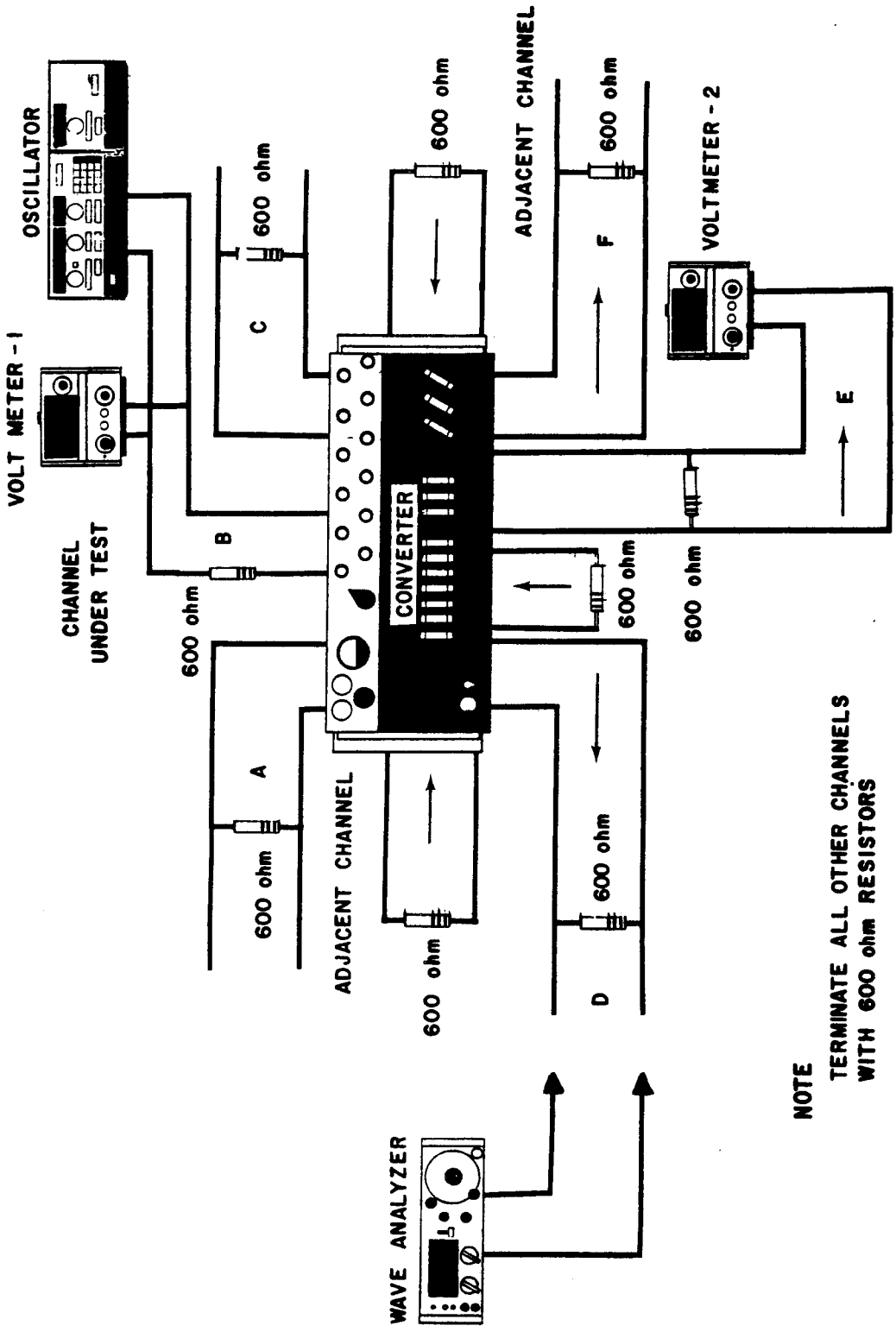



Figure 10. Crosstalk test setup.



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b. 2-wire to 4-wire configuration

(1) Set the audio signal generator (OSC) to 300 Hz at 0.98 Vrms, as read on voltmeter VM-1 and record the output on voltmeter VM-2.

(2) Repeat step (1) for test frequencies of 500, 1000, 1500, 2000, 2500, 3000, and 3500 Hz and 5, 10, 20, 30, 40, 50, 75 and 100 kHz.

c. 4-wire to 2-wire configuration

(1) Repeat steps b(1) and b(2).

5.1.15 Noise Level

a. Set up the test as shown in figure 12, page 26, with a randomly selected converter channel for test.

b. Set the NMS to FIA weighting and record the level.

c. Connect the NMS to the converter 2-wire port and record the level.

5.1.16 Systems Compatibility

NOTE: The objective of the test is to see if the signal converter under test is compatible with existing wired communication equipment. Each channel of the converter should accept/extend calls from/to 4-wire automatic switchboards.

a. Select a channel for test and set up the equipment as shown in figure 13, page 27, with the power supply set at the appropriate input voltage.

(1) Activate the SRI for the CUT and the night alarm by removing the handset (TA-341).

(2) Connect another handset (TA-312/TA-236) to the converter 2-wire port and the SRI should be deactivated.

(3) Remove the handset (TA-312) and establish two-way voice communication between the two phases.

b. Set up the test signal converter as illustrated in figure 14, page 28.

(1) Remove one handset (TA-341) and establish two-way voice communication through a switchboard (SB-3614) with another handset (TA-341).

(2) Repeat set (1) through another switchboard (SB-22) to another type phone (TA-312/TA-236).



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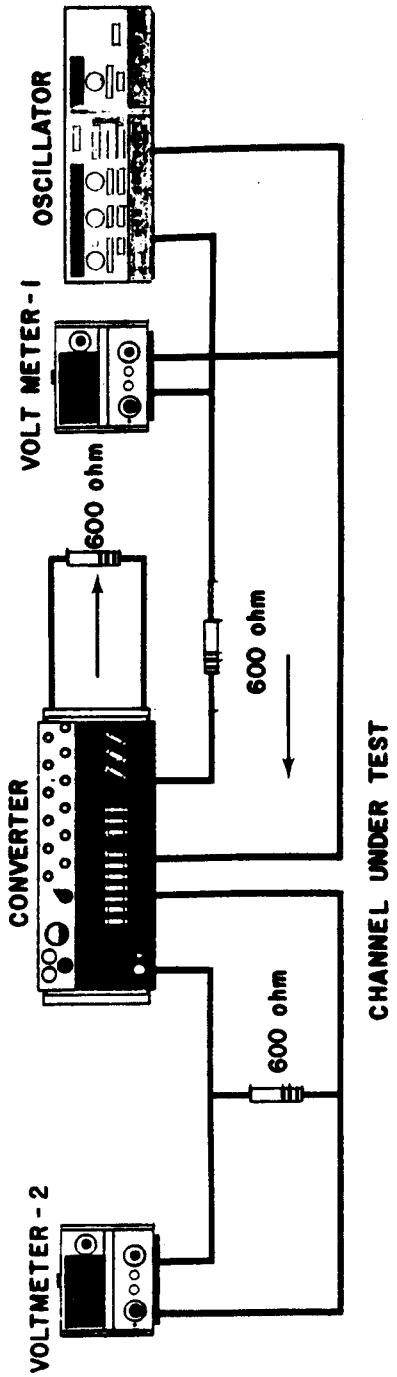
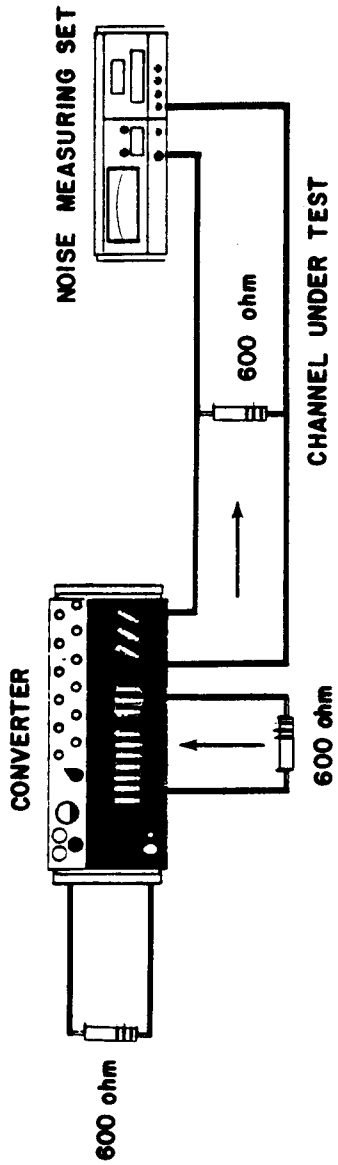



Figure 11. Insertion loss/frequency response test setup.



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NOTE

TERMINATE ALL OTHER CHANNELS WITH 600 ohm RESISTORS

Figure 12. Noise level test setup.



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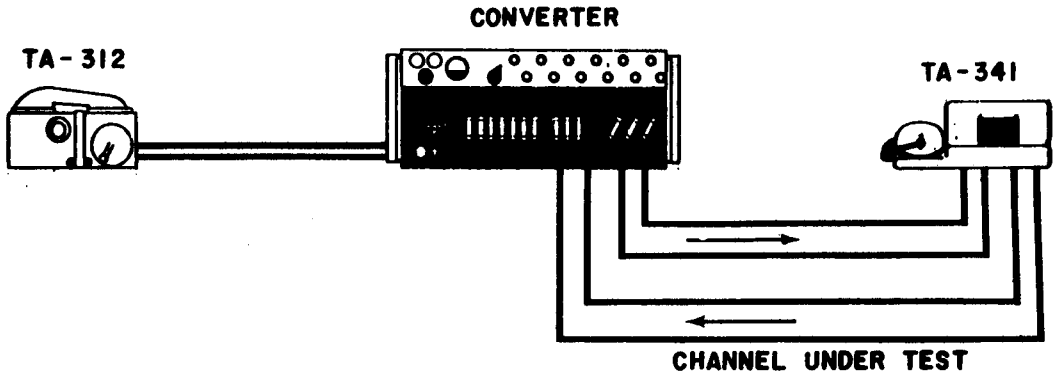



Figure 13. Two-wire to four-wire test setup.



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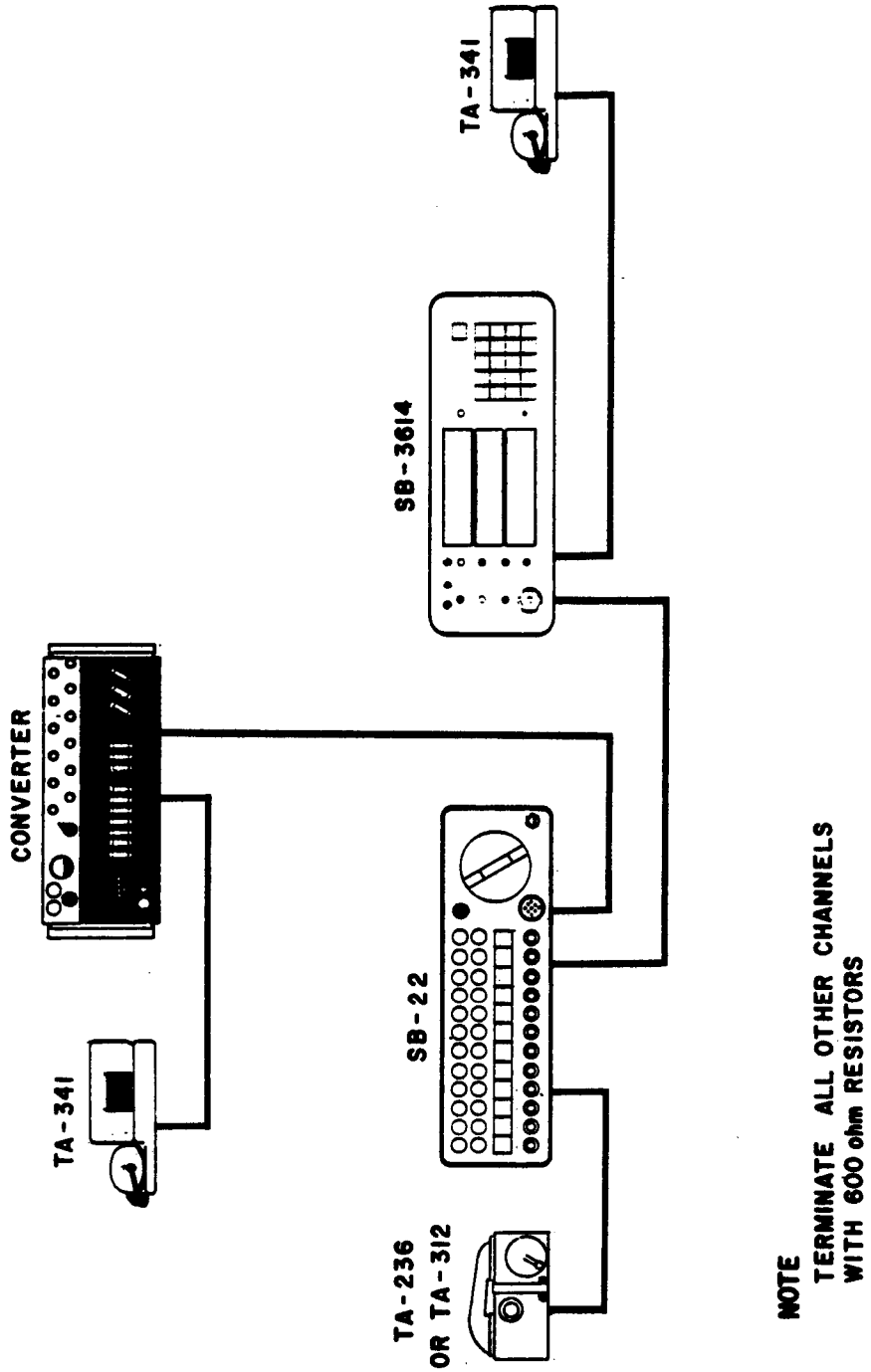



Figure 14. Systems compatibility test setup.



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## 5.2 Data Required

### 5.2.1 Preparation

Assemble records of the following data recorded prior to testing:

a. Nomenclature, serial number(s), manufacturer's name, and function of the item(s) under test.

b. Nomenclature, serial number, accuracy tolerances, calibration requirements, and last date calibrated of the test equipment selected for the tests.

c. Appropriate photographs of the test item.

### 5.2.2 Test Data

a. Record throughout the test in the project log book in chronological order pertinent remarks and observations which will augment test data and support engineering evaluation and analysis of the technical performance of the test item.

b. Acquire throughout the test supporting photographs, calibration records and recordings of test anomalies and note deviations from the test plan found necessary.

#### 5.2.2.1 Record Signaling Characteristics (Incoming Calls) data as follows:

a. Channel under test
b. OSC frequency
c. OSC amplitude
d. Output frequency
e. Output amplitude
f. Delay/duration time
g. Verification of channel operation

#### 5.2.2.2 Record Signaling Characteristics (Outgoing Calls) data as follows:

a. Channel under test
b. Output frequency
c. Output amplitude
d. OSC frequency
e. OSC output level
f. Condition of SRI

#### 5.2.2.3 Record Signaling Characteristics (Release Transmission) as follows:

a. Channel under test
b. Frequency
c. Output level
d. Condition of channel lamps
e. Presence and duration of 2600 and 570 Hz signals

#### 5.2.2.4 Record Audible Alarm as follows:

a. Channel under test
b. Presence of seize signal
c. Presence of release signal
d. Seize time
e. Release time
f. Presence and adjustability of alarm sound



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5.2.2.5 Record Tone Detectors data as follows:

a. Channel under test
b. OSC frequency (See Table 1)
c. OSC output level
d. Converter detector response time

5.2.2.6 Record Detection data as follows:

a. Channel under test
b. OSC frequency
c. OSC output in dB
d. Condition of all channel lamps
e. Presence and duration of 570 Hz release acknowledge signal

5.2.2.7 Record Return Loss data as follows:

a. Channel under test
b. OSC frequency
c. OSC output level
d. Return loss (dB) for each test frequency

5.2.2.8 Record Transmission Loss data as follows:

a. Channel under test
b. OSC frequency (Hz)
c. OSC waveform amplitude (Vrms)
d. Channel output level (dBm) for each input signal

5.2.2.9 Record 20 Hz Rejection data as follows:

a. Channel under test
b. OSC frequency (Hz)
c. OSC amplitude (Vrms)
d. Channel output level (dBm) for each input signal

5.2.2.10 Record Longitudinal Balance data as follows:

a. Channel under test
b. WA-OSC test frequency
c. WA-OSC output level
d. VM reading (dB) for each SW-1 setting
e. Calculated longitudinal balance

5.2.2.11 Record Harmonic Distortion data follows:

a. Channel under test
b. OSC frequency (Hz)
c. Level (dBm) of the first through tenth harmonic for each fundamental

5.2.2.12 Record Hybrid/Transhybrid Loss data as follows:

a. Channel under test
b. OSC test frequency (Hz)
c. OSC amplitude (Vrms)
d. Output level (dBm)



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5.2.2.13 Record Crosstalk data as follows:

a. Channel under test
b. OSC test frequency (Hz)
c. OSC amplitude (Vrms)
d. WA outpt level (dBm)
d. Channel output level (dBm)

5.2.2.14 Record Frequency Response data as follows:

a. Channel under test
b. OSC frequency (Hz)
c. OSC amplitude (Vrms)
d. Output level (dBm) for each input signal

5.2.2.15 Record Noise Level data as follows:

a. Channel under test
b. Noise level (dBm)

5.2.2.16 Record System Compatibility data as follows:

a. Channel under test
b. Condition of SRI
c. Delay in completing call
d. Verification of operation

## 6.0 DATA REDUCTION AND PRESENTATION

### 6.1 General

Processing of raw test data, in general, includes but is not limited to the following steps:

a. Marking test data for identification and correlation according to subtest.

b. Organizing data into tabular and graphical form.

c. Modifying data to correct for nonstandard conditions.

d. Determining the statistical variation of the results in terms of the average value and standard deviation of the particular quantities and the correlation among two or more quantities.

NOTE: The test directive (or specification) itself serves to define the types and characteristics of the raw test data, and the ultimate objective of the test program defines the form of the test data desired.

Specific instructions for the reduction and presentation of individual subtest data are outlined in subsequent paragraphs.



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6.2 Subtest Data

a. The recorded data of subtests 5.1.1, 5.1.2, 5.1.3, 5.1.4, 5.1.5, 5.1.6, 5.1.7, 5.1.8, 5.1.9, 5.1.11, 5.1.12, 5.1.15, 5.1.16 will be compared with specified requirements to determine if any variation exist. Any malfunction or unacceptable performance will be analyzed to determine if it is characteristic of the test item or an isolated case. If the incident is found to be characteristic of the test item, an evaluation will be made to determine the input on the overall system performance and what corrective actions are required.

b. The recorded longitudinal balance data of subtest 5.1.10 and the minimum allowable longitudinal balance will be plotted to ascertain compliance with specified requirements. Any difference will be evaluated for the effect on system performance.

c. The recorded crosstalk data of subtest 5.1.13 and the specified maximum allowable crosstalk level will be tabulated and plotted to permit analysis for ascertaining compliance with the specification requirements.

Any differences between test item data and specifications must be evaluated for its effect on system performance.

d. The recorded frequency response data of subtest 5.1.14 and the specified frequency will be tabulated and plotted to permit analysis for ascertaining compliance with the specification requirements. And deviations will be evaluated for its impact on system performance.

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## APPENDIX A

### PERFORMANCE TEST CHECKLIST

| | YES | NO | N/A | DATE |
|---|---|---|---|---|
| 1. Test item documentation received | ___ | ___ | ___ | _____ |
| 2. Test item availability schedule determined | ___ | ___ | ___ | _____ |
| 3. Test procedures defined | ___ | ___ | ___ | _____ |
| Subtest 1 | ___ | ___ | ___ | _____ |
| Subtest 2 | ___ | ___ | ___ | _____ |
| Subtest 3 | ___ | ___ | ___ | _____ |
| 4. Required facilities determined and scheduled | | | | |
| Subtest 1 | ___ | ___ | ___ | _____ |
| Subtest 2 | ___ | ___ | ___ | _____ |
| Subtest 3 | ___ | ___ | ___ | _____ |
| 5. Required instrumentation determined and scheduled | | | | |
| Subtest 1 | ___ | ___ | ___ | _____ |
| Subtest 2 | ___ | ___ | ___ | _____ |
| Subtest 3 | ___ | ___ | ___ | _____ |
| 6. Required test personnel determined and scheduled | | | | |
| Subtest 1 | ___ | ___ | ___ | _____ |
| Subtest 2 | ___ | ___ | ___ | _____ |
| Subtest 3 | ___ | ___ | ___ | _____ |
| 7. Conduct necessary aspects of safety checklist | ___ | ___ | ___ | _____ |
| 8. Conduct necessary aspects of MTSP checklist | ___ | ___ | ___ | _____ |
| 9. Conduct necessary aspects of photographic checklist | ___ | ___ | ___ | _____ |
| 10. Review testing priority checklist | ___ | ___ | ___ | _____ |
| a. All items received | ___ | ___ | ___ | _____ |
| b. Test schedules | ___ | ___ | ___ | _____ |
| (1) Prepared | ___ | ___ | ___ | _____ |
| (2) Coordinated | ___ | ___ | ___ | _____ |
| (3) Submitted | ___ | ___ | ___ | _____ |



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| | YES | NO | N/A | DATE |
|---|---|---|---|---|
| c. Procedure developed for each operation | ___ | ___ | ___ | _____ |
| d. Test site(s) inspected | ___ | ___ | ___ | _____ |
| e. Safety hazards or critical problems solved | ___ | ___ | ___ | _____ |
| f. All pretests operations completed | ___ | ___ | ___ | _____ |
| g. Item prepared for test | ___ | ___ | ___ | _____ |
| h. SOPs and checklist posted at site(s) | ___ | ___ | ___ | _____ |
| i. Test team briefed | ___ | ___ | ___ | _____ |
| j. Required capabilities on hand | ___ | ___ | ___ | _____ |
| 11. Initiate testing per approved test plan and schedule established | ___ | ___ | ___ | _____ |
| 12. Use testing checklist developed for each subtest | ___ | ___ | ___ | _____ |
| 13. Submit equipment performance reports | ___ | ___ | ___ | _____ |
| 14. Prepare memorandums for record to document changes in test plans, directives, or procedures resulting from conferences, telephone calls or conversations | ___ | ___ | ___ | _____ |
| 15. Insure that found changes in or out of the project have documented authorization | ___ | ___ | ___ | Ongoing |
| 16. Receive internal reports written by supporting organizations | ___ | ___ | ___ | Ongoing |
| 17. Begin early preparation of report paragraphs including subtest-by-subtest writings while subsequent subtests are in progress | ___ | ___ | ___ | _____ |
| 18. Continually monitor expenditure of funds using ITRMS and work planning and control reports. Request additional funds as required | ___ | ___ | ___ | Ongoing |
| 19. Begin data reduction early and continue reduction and inspection continuously throughout the test | ___ | ___ | ___ | _____ |



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| | | YES | NO | N/A | DATE |
|---|---|---|---|---|---|
| 20. | Make notes during the conduct of tests regarding inadequacies in test techniques, facilities, or instrumentation which served to delay test completion, increase costs or compromise test results | ___ | ___ | ___ | Ongoing |
| 21. | Submit delays when schedule cannot be met | ___ | ___ | ___ | Ongoing |
| 22. | If funds are becoming exhausted. Suspension of the test program should be coordinated with HQ, TECOM. | ___ | ___ | ___ | Ongoing |
| 23. | Report problem areas on a daily basis to supervisor | ___ | ___ | ___ | Ongoing |
| 24. | Make entries to project file and project log as appropriate | ___ | ___ | ___ | Ongoing |
| 25. | Conduct test activity EPR review as necessary | ___ | ___ | ___ | _____ |
| 26. | All subtests complete | ___ | ___ | ___ | _____ |
| 27. | Analyze data | ___ | ___ | ___ | _____ |
| 28. | Prepare test report | ___ | ___ | ___ | _____ |



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# APPENDIX B

## Signal Converter Data Sheet

### Performance Test

1. Test Item(s):_____ 2. SN:_____

3. Test Sequence No:_____ 4. Test Operator(s):_____

5. Date:_____ 6. Time:_____

7. Test Procedure Function:_____ 8. Channel ID:_____

| Input | | | Response | | | | |
|---|---|---|---|---|---|---|---|
| Frequency | Level | Other | Frequency | Level | Duration | Delay | Other | Remarks |



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## APPENDIX B

## CONVERTER GENERATED SIGNALS CHECKLIST

| Function | Freq. in Hz | Single Tone Level in dBM | Duration | Remarks |
|---|---|---|---|---|
| Digits 1 | 697/1209 | -14± 2 | Operator Controlled | Operator originated |
| 2 | 697/1336 | | | |
| 3 | 697/1477 | | | |
| 4 | 770/1209 | | | |
| 5 | 770/1336 | | | |
| 6 | 770/1477 | | | |
| 7 | 852/1209 | | | |
| 8 | 852/1336 | | | |
| 9 | 852/1477 | | | |
| 0 | 941/1366 | | | |
| Seize | 2250 | -14± 2 | Operator Controlled | Held until seize ack. is heard |
| Seize acknowledge | 570 | -14± 2 | Continuous until 250 ms after seize stops | |
| Ringback tone | 425 | -14± 2 | Until operator answers | 2 s on 4 s off |
| Conference | 941/1477 | -14± 2 | Operator Controlled | Operator originated |
| Recall | 941/1209 | -14± 2 | Operator Controlled | Operator originated |
| Release | 2600 | -14± 2 | Continuous up to 3 to 10 s | Operator originated. Shutoff by release ack. |
| Release acknowledge | 570 | -14± 2 | | Cutoff 250 ms after release ends |



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# APPENDIX C

## SIGNAL CONVERTERS

The many types and possible combinations of communication facilities, systems, and associated signaling schemes comprising a military network requires the use of signal converters in different forms at strategic points in order to effect overall signaling compatibility.

These items principal purpose is to alter the form of telephone to telegraph functional signals in circuits having dissimilar line or terminal signaling characteristics. Functional signals are defined as those employed in most conjunction circuits to operate signaling devices or switching equipment as opposed to the intelligence-bearing signals (speech, data, and images).

The majority of communication circuits employ two-way signaling, therefore signal converters which may be required must be capable of performing the basic signaling functions in both directions regardless of the circuit configuration such as 2-wire, 4-wire, or combinations of the two in tandem. See figure 1 and 2 for a basic representation circuit. The basic functions and characteristics of a signal converter are as follows:

a. Accept signaling information in one electrical form form one direction and relay it in a different electrical form such as DC to AC, and low-frequency AC to a high-frequency AC.

b. Accurately repeat the signaling condition and sequence such as "on hook/off hook", telephone dial pulses to "touch tone" frequencies.

c. Provide an essentially loss-less and distortion-less voice path through the converter when in the "talking" condition.



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## APPENDIX D

## REFERENCES


| | | |
|---|---|---|
| 1. | MIL-STD-188-100 | Common Long Haul and Tactical Communication System Technical Standards, 15 November 1972 |
| 2. | MIL-STD-454F | General Requirements for Electrical Equipment, 15 March 1978 |
| 3. | TM11-678 | Fundamentals of Telephony |
| 4. | TM11-486 | Electrical Communication Systems Engineering |
| 5. | Welch, Samuel | Signaling in Telecommunications Networks, The Institution of Electrical Engineers, 1979 |
| 6. | ITT | Reference Data for Radio Engineers, 1977 |
| 7. | TOP 6-2-055 | Communication Security Equipment |
| 8. | TOP 6-2-065 | Data Transmission Equipment |
| 9. | TOP 6-2-110 | Handset, Telephone |
| 10. | TOP 6-2-200 | PCM Multiplexers |
| 11. | TOP 6-2-230 | Radio Control Equipment |
| 12. | TOP 6-2-265 | Switchboards, Manual |
| 13. | TOP 6-2-290 | Terminals, Telephone and Telegraphs |
| 14. | TECOM PAM 310-4 | Index of Test Operations Procedures |
| 15. | TECOM Supplement 1 w/change 1, to AMC Reg 310-6 | Quality Assurance Publication |
| 16. | TECOM Pam 70-3 | Project Engineers' Handbook |
| 17. | TECOM Reg 70-24 | Documenting TECOM Testing |
| 18. | DARCOM/AMCR 700-38 | Equipment Performance Report |




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# APPENDIX E

## ABBREVIATIONS

| | |
|---|---|
| AFC | automatic frequency control |
| ATT | attenuator |
| CUT | Channel/circuit under test |
| dB | decibel |
| dBm | Power in dB relative to 1 mW |
| DTMF | dual tone multi-frequency |
| EC | electronic counter |
| Hz | Hertz |
| LB | longitudinal balance |
| ms | millisecond |
| NMS | noise measuring set |
| OSC | oscillator |
| rms | root mean square |
| s | second |
| SB | switchboard |
| SIDTC | Single Intergrated Development Test Cycle |
| SRI | service request indicator |
| SW | switch |
| Vrms | voltage root mean square |
| VM | voltmeter |
| WA | wave analyzer |



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